Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3341-WE01

Title: (BEng/MEng)

Environmental Engineering 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

(a) Apart from temperature and pressure, what are the two most important parameters governing whether or not an element is 'available' to the soil microbiome? Give an example of how a change in these two parameters might affect soil health.

[25%]

(b) Explain why soil health is important to engineers and describe how at least 2 UN Sustainable Development Goals are directly linked to soil health and potentially each other.

[25%]

(c) Explain the physical, chemical and biological processes that are operating in a sewage treatment plant (in terms of inorganic and organic treatment) at primary, secondary (at least two options) and tertiary stages making sure you discuss the pros and cons of each process. Give details of what legislation is being adhered to by the water companies in ensuring water quality compliance.

[50%]

continued

Q2

(a) In the following reaction, what is being oxidised and what is being reduced? What does the equation represent?

$$C_6H_{12}O_6 + O_2 \longrightarrow 6\,CO_2 + 6\,H_2O \tag{1}$$

[10%]

(b) Which elements are the main constituents of BOD and COD and explain what is the difference between them and why this might be a problem for sewage treatment works? Would the $C_6H_{12}O_6$ molecule in Eq. (1) above contribute to BOD or COD?

[20%]

(c) Explain how and why, using diagrams where useful, an oil spill on a sandy soil would be easier to clean up than an oil spill on a peaty soil and what sort initial soil conditions might affect the severity of the contamination event. Also describe the most appropriate remediation methods you might use to clean up this contamination in both soil and groundwater.

[70%]

continued

Q3

(a) Use the Airy wave equation to derive an expression relating shallow-water wavelength to deep-water wavelength and still-water depth.

[**30%**]

(b) Use your previous result and the wave power equation to derive an expression relating shallow-water wave-height to deep-water wave-height and still-water depth.

[**30%**]

(c) A sloped concrete coastal defence is required to ensure wave breaking and to prevent 98% of incoming waves from overtopping. The still-water depth is 3 m. The deep-water wavelength is 30 m. The local wave-height is 2 m. Specify, with supporting calculations, a suitable slope and crest elevation above seabed.

[**40%**]

continued

ENGI3341-WE01 Environmental Engineering 3
Formula sheet

Airy wave equation:

$$L = \frac{gT^2}{2\pi}\tanh\left(\frac{2\pi\eta_0}{L}\right)$$

Wave-power equation :

$$R = \frac{\rho g L H^2}{16T}\left[1 + \frac{4\pi\eta_0}{L}\mathrm{csch}\left(\frac{4\pi\eta_0}{L}\right)\right]$$

Iribarren number:

$$\xi_p = \frac{\tan\beta}{\sqrt{H/L_0}}$$

Wave run-up for concrete slope:

$$R_{u2} = \begin{cases} 1.6\xi_p, & \xi_p < 2.5 \\ 4.5 - 0.2\xi_p, & \xi_p > 2.5 \end{cases}$$

Wave breaking:

$$\xi_p < 2$$

Useful limits and asymptotics:

$$\lim_{x\to\infty}\tanh x = 1$$

$$\tanh x = x + O(x^3)$$

$$\lim_{x\to\infty}\mathrm{csch} x = 0$$

END